MIL-L-25242B(USAF)
19 September 1958
Superseding
MIL-L-25242A(USAF)
24 January 1957

MILITARY SPECIFICATION

LOADING DEVICE, PRACTICE BOMB, TYPE MD-1

1. SCOPE.- This specification establishes the requirements for one type of equipment designated as Loading Device, Practice Bomb, Type MD-1 for use in loading practice bombs BDU-2/B, MB-2, or MK-76 into the Trainer, Special Weapons Device, Type MN-1, and Type MN-1A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-T-9107	Test Reports, Preparation Of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

405 Storage Life-Aeronautical Articles

DRAWINGS

Republic Aviation Corporation

GMD-52-30100 Loading Device, Practice Bomb, Type MD-1

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION TESTING.- This specification makes provisions for preproduction testing.

3.2 DESIGN AND CONSTRUCTION.- The design and construction of the Loading Device, Practice Bomb, Type MD-1 shall be in accordance with Republic Aviation Corporation Drawing GMD-52-30100 and its associated detail drawings.

3.3 PERFORMANCE.- The loading device shall perform satisfactorily its function of loading the following practice bombs into position in the Trainer, Special Weapons Device, Type MN-1 and Type MN-1A:

a. Bomb, Practice BDU-2/B
b. Bomb, Practice, Target, Miniature, Type MB-2
c. Bomb, Practice, 25 Pound, MK-76 Mod 2

3.4 IDENTIFICATION OF PRODUCT.- The loading device shall be marked for identification in accordance with MIL-STD-130.

3.5 WORKMANSHIP.- The loading device shall be fabricated and finished in a manner such that criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria must apply equally to all accessories and to all external non-functioning parts (e.g., cases, cover plates, nameplates) as well as to functional parts. Particular attention shall be given to neatness, marking of parts and assemblies, plating, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

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3.6 GOVERNMENT LOANED PROPERTY.- The following items of equipment shall be loaned to the contractor, upon his request, as government loaned property, for the purpose of conducting functional tests on the loading device:

Item No.	Stock No.	Part No.	Item Description	Quantity
1	0602-AAF 873000		Trainer, Special Weapons Device, Type MN-1	1 ea
			or	
1	1115-610-0451		Trainer, Special Weapons Device, Type MN-1A	1 ea
2			Bomb, Practice BDU-2/B	2 ea
			or	
2	1115-554-4517		Bomb, Practice, Target, Miniature, Type MB-2	2 ea
			or	
2	1325-529-7566		Bomb, Practice, 25 Pound, MK-76 Mod 2	2 ea

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the Loading Device, Type MD-1, shall be classified as follows:

a. Preproduction Tests . . . . . See 4.2
b. Acceptance Tests. . . . . See 4.3

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES.- The preproduction test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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4.2.2 PREPRODUCTION TESTS.- Preproduction tests shall consist of all tests described herein under "4.5 TEST METHODS" of this specification.

4.3 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual Tests
b. Sampling Plan and Tests

4.3.1 INDIVIDUAL TESTS.- Each item shall be subjected to the following tests as described under "4.5 TEST METHODS" of this specification:

a. Examination of Product
b. Conformance to Drawings
c. Adjustment of Torque Limiter

4.3.2 SAMPLING PLAN AND TESTS

4.3.2.1 PERIODIC SAMPLING PLAN.- One Loading Device, Practice Bomb, Type MD-1, shall be selected at random from each 50 or fraction thereof produced and subjected to the tests listed below as described under "4.5 TEST METHODS" of this specification:

Functional Test

4.3.2.1.1 REJECTION AND RETEST.- When the sample item fails to meet the specified test, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.3.2.1.2 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of this specification.

4.3.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS.- The test conditions are described under the individual tests to which they apply. See 4.5.

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4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT.- Each loading device shall be examined to determine that the workmanship meets the requirements of this specification.

4.5.2 CONFORMANCE TO DRAWINGS.- Each loading device shall be inspected for strict conformance to the pertinent drawings and specifications.

4.5.3 ADJUSTMENT OF TORQUE LIMITER.- On each loading device, the "Torque Limiter #330-1" shall be adjusted to slip when output force of the saddle reaches 750 pounds $\pm$ 50 pounds. If this adjustment cannot be made on any loading device, the loading device shall be rejected.

4.5.4 FUNCTIONAL TEST

4.5.4.1 TEST PROCEDURE

a. Adjust "Torque Limiter #330-1" to slip when output force of saddle reaches 750 pounds $\pm$ 50 pounds.

b. Mount loading device to side of MN-1 Trainer or MN-1A Trainer.

c. Insert practice bomb (BDU-2/B, MB-2, or MK-76 Mod 2) on loading saddle, and crank loading device until bomb is elevated and locked in position.

d. Crank loading saddle down to initial position.

e. Slide gear box and saddle to second position.

f. Insert practice bomb on loading saddle, and crank loading device until bomb is elevated and locked into position.

NOTE: If the torque limiter is set too low, the loading device clutch will slip before the bomb is loaded. If it is set too high, the clutch will not slip when the operation is continued after the bomb is loaded into position. The proper setting must be used in order to obtain the correct amount of pressure to avoid injuring the ejection mechanism.

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4.5.5 LIFE CYCLE TEST

4.5.5.1 TEST PROCEDURE

a. Adjust "Torque Limiter #330-1" to slip when output force of saddle reaches 750 pounds ± 50 pounds.

b. Mount loading device to side of MN-1 or MN-1A Trainer, or to a suitable test frame. If a test frame is used, a restraining plate shall be located at the same elevation above the loading device as is the MN-1 ejector plunger in its fully retracted position. This restraining plate shall be capable of resisting repeated forces in excess of 800 pounds.

c. Insert practice bomb (BDU-2/B, MB-2, or MK-76 Mod 2) on loading saddle.

d. Crank loading device until bomb is raised to proper elevation. If the MN-1 or MN-1A Trainer is used, provision shall be made to prevent the bomb from locking into position.

e. Crank loading saddle down to initial position.

f. Repeat steps "d" and "e" 1000 times.

g. Slide gear box and saddle to second position.

h. Repeat steps "d" and "e" 1000 times.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVELS A AND C

5.1.1.1 DISASSEMBLY.- The Loading Device, Practice Bomb, Type MD-1 shall be disassembled as far as practical in order to reduce cube.

5.1.1.2 PRESERVATION

5.1.1.2.1 GEAR ASSEMBLY.- The gears and gear housing shall be preserved in accordance with method 1A, MIL-P-116, with a contact preservative.

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5.1.1.2.2 LOADING DEVICE EXCLUDING GEAR ASSEMBLY.- The loading device, excluding the gear assembly, shall be preserved in accordance with method I of MIL-P-116.

5.1.1.3 PACKAGING.- The loading device shall be packaged according to the manufacturer's commercial practice. Each package shall contain one complete loading device, disassembled in accordance with 5.1.1.1, and preserved in accordance with 5.1.1.2.

5.2 LEVELS OF PACKING

5.2.1 LEVEL A.- Loading devices preserved and packaged to meet 5.1 shall be packed in exterior type shipping containers meeting PPP-B-601 or PPP-B-621. As far as practicable, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds.

5.2.2 LEVEL C.- Loading devices shall be packed as for level A. In addition, packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules, or regulations of other common carriers as applicable to the mode of transportation.

5.3 PHYSICAL PROTECTION.- Cushioning, blocking, bracing and bolting as required shall be in accordance with JAN-P-100 except that for domestic shipments, water-proofing requirements for cushioning materials and containers shall be waived.

5.4 MARKING FOR SHIPMENT AND STORAGE.- Interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

Loading Device, Practice Bomb, Type MD-1

5.4.1 The date of marking shall be the same for interior and exterior containers.

5.4.2 REINSPECTION DATE MARKING.- Reinspection markings shall be added in accordance with ANA Bulletin 405.

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6. NOTES

6.1 INTENDED USE.- The Loading Device, Practice Bomb, Type MD-1 shall be used for the purpose of loading the following practice bombs into the Trainer, Special Weapons Device, Type MN-1, or into the Trainer, Special Weapons Device, Type MN-1A:

a. Bomb, Practice BDU-2/B
b. Bomb, Practice, Target, Miniature, Type MB-2
c. Bomb, Practice, 25 Pound, MK-76 Mod 2

6.2 ORDERING DATA.- Invitations for bids, contracts and purchase orders should state the conditions for the following:

6.2.1 PREPRODUCTION SAMPLES.- It is expected that the contract or purchase order will specify that three units representative of the production loading device will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order, should specify the point of inspection for these tests.

6.3 SOURCES.- When requesting publications, refer to both title and number.

6.4 DEFINITIONS

a. PREPRODUCTION TESTS.- Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove, prior to his starting production, that the loading device meets the requirements of this specification. Any production prior to the approval of the preproduction samples shall be at the contractor's risk.

b. ACCEPTANCE TESTS.- Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

c. AS SPECIFIED HEREIN.- The expression "as specified herein" refers to all sections and subsections of this specification where the associated referenced requirement has a logical engineering application and dependence, and is not necessarily limited to the subsection wherein the expression is used.

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